

Inho Yoo

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Portfolio: <https://inhotwin-png.github.io/inhoyoo.github.io/>

EDUCATION

Kwansei Gakuin University | Hyogo, Japan

Bachelor of Science in Physics and Astronomy | *Expected March 2027*

- **Advisor / Senior Thesis Supervisor:** Prof. Shuji Matsuura
- **Relevant Coursework:** Astrophysics, Infrared Astronomy, X-ray Astronomy, Computational Physics

RESEARCH INTERESTS

Planetary Surface Composition • Infrared and X-ray Remote Sensing • Scientific Machine Learning • Space-Resource Exploration

RESEARCH EXPERIENCE

Undergraduate Researcher, Matsuura Laboratory —
Kwansei Gakuin University

Project: *Spatial Distribution Analysis of the SPLICES Target Catalog*

- Examined the SPLICES target catalog, a 3.1 GB FITS dataset containing ~9 million background sources selected for studies of interstellar ice.
- Used Python and Astropy's memory-mapped FITS access to randomly sample 10,000 sources and visualize their right ascension and declination.
- Examined the sampled targets' sky distribution, including the Galactic-plane pattern in equatorial coordinates.
- Planned senior thesis (not yet started): investigate lunar composition using infrared and X-ray observations, returned-sample measurements, and machine learning. Initial work will assess data availability, spatial correspondence, and sample suitability for evaluation.

SELECTED RESEARCH PROJECT

Multimodal Compositional Inference for Space-Resource Exploration — *Research Proposal*

- Proposed an infrared and X-ray machine-learning framework to assess planetary composition and prioritize targets for future space-resource exploration; not yet implemented or validated.
- Outlined a possible Moon → Bennu → Ryugu progression to study complementary, noisy, and missing observations, subject to data availability and suitability.
- Proposed comparisons with returned-sample measurements, accounting for spatial correspondence and sample representativeness when evaluating remote composition estimates.

TECHNICAL SKILLS

Programming: Python | **Scientific Computing:** Astropy, NumPy, Pandas, Matplotlib, PDS4 Tools
Data Analysis: FITS catalog handling, memory-mapped access, random sampling, coordinate visualization
Tools: LaTeX, Visual Studio Code | **Languages:** Korean, Japanese, and English — Fluent